



AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE

Alex Johnson

FINAL SCORE

18 / 70

weighted points

CORRECT ANSWERS

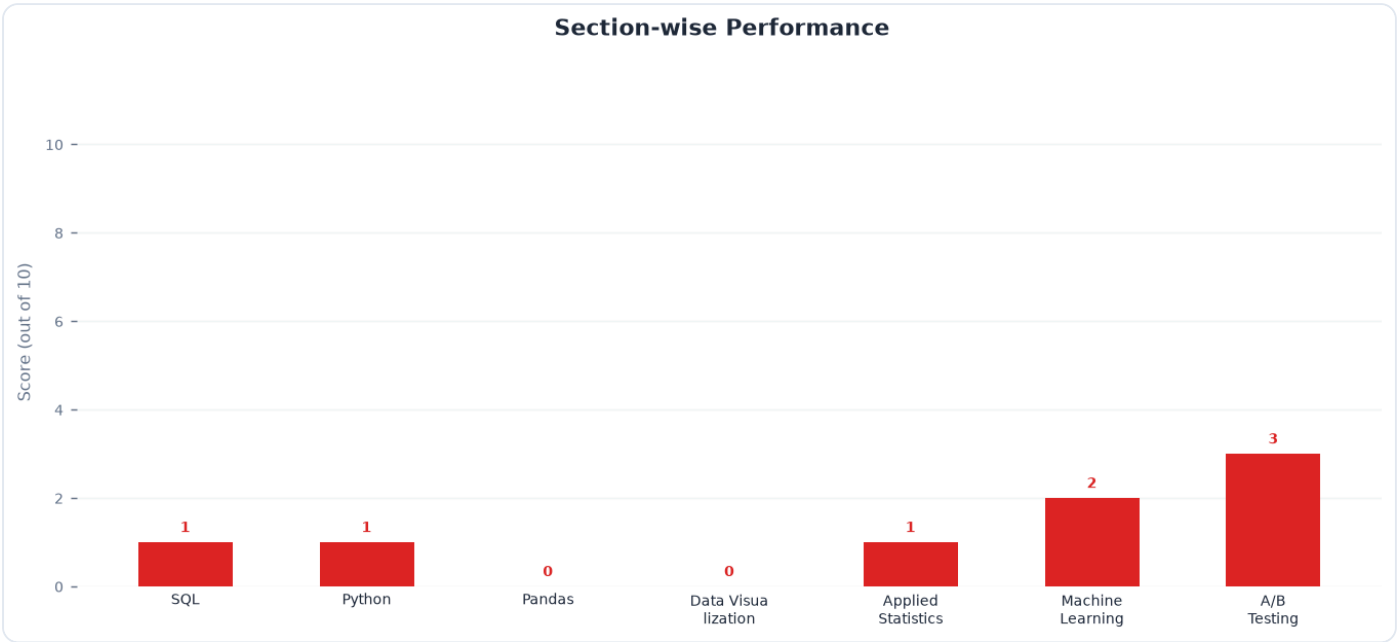
8 / 35

questions

NEEDS IMPROVEMENT

Assessment Date: July 25, 2026 Report Generated: July 25, 2026 at 16:40 UTC Role: Data Science Expert

OVERALL RATING	SUCCESS RATE	WEIGHTED SCORE	TOP SKILL	NEEDS IMMEDIATE ATTENTION
★☆☆☆☆	25.7%	18 / 70	A/B Testing	Pandas



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	1 / 10	6.8	-5.8	Critical Gap
Python	1 / 10	6.2	-5.2	Critical Gap
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	0 / 10	6.0	-6.0	Critical Gap
Applied Statistics	1 / 10	5.5	-4.5	Critical Gap
Machine Learning	2 / 10	5.2	-3.2	Needs Work
A/B Testing	3 / 10	6.4	-3.4	Needs Work

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	1	4	0	20%
Python	5	5	1	4	0	20%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	0	5	0	0%
Applied Statistics	5	5	1	4	0	20%
Machine Learning	5	5	2	3	0	40%
A/B Testing	5	5	3	2	0	60%

Executive Summary

The candidate achieved a correct-answer count of 8 out of 35 (22.9% accuracy), resulting in a weighted score of 18 out of 70. This performance indicates systemic gaps across foundational data science domains, particularly in query optimization, data manipulation mechanics, and statistical diagnostics. Immediate, structured technical upskilling is required to meet the baseline expectations for professional data science roles.

Key Strengths

- Conceptual Python Awareness:** Demonstrates familiarity with advanced object-oriented Python concepts, including descriptors and custom attribute management.
- SQL Join Logic:** Understands relational database schemas and the logical utility of self-joins to resolve multi-condition queries.

- **Pandas Pipeline Exposure:** Shows comfort with chaining Pandas methods (e.g., query, assign, groupby) to perform multi-step data transformations.

Top Blind Spots

- **SQL Aggregations and CTEs:** The candidate struggles with conditional aggregations (such as using `CASE WHEN` inside `COUNT`) and lacks knowledge of loop-prevention techniques in recursive CTEs.
- **Pandas Indexing and Windowing:** There is a critical misunderstanding of how non-monotonic indexes break rolling window calculations and how label-based slicing behaves on MultiIndexes.
- **Statistical and ML Diagnostics:** The candidate misinterprets essential diagnostic plots, specifically confusing heavy-tailed symmetric distributions with skewness on Q-Q plots, and high-bias underfitting with optimal model fits on learning curves.

Actionable Learning Resources

SQL Optimization: Complete the "Advanced SQL" and "Window Functions" interactive tracks on the Mode Analytics SQL Tutorial platform to master conditional aggregation and performance-tuning.

✓ Action Item

Pandas Mechanics: Read Chapter 3 ("Data Manipulation with Pandas") of Jake VanderPlas's *Python Data Science Handbook*, focusing specifically on MultiIndex slicing and monotonic index requirements for rolling windows.

✓ Action Item

Statistical Diagnostics: Study Chapter 2 ("Statistical Learning") and Chapter 7 ("Moving Beyond Linearity") of *An Introduction to Statistical Learning (ISLR)* to master Q-Q plot interpretation, bias-variance tradeoffs, and model calibration.

✓ Action Item

Python Protocols: Work through the "Descriptor HowTo Guide" in the official Python Documentation to master the dunder protocol (`__get__`, `__set__`, `__delete__`).

✓ Action Item